PeriGen, Inc.

PeriWatch® Tracings System Requirements

PeriWatch® Tracings 2019 Rev 1.02

PeriGen
Advanced
Perinatal Systems

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Standard / Enterprise Model (Multi-Facilities) Server Requirements

- A full enterprise implementation of PeriWatch Tracings and Decision Support modules requires a base configuration of 3 server VMs as follows:
  - PeriWatch Tracings Application/DB/HL7 Interface VM
  - PeriWatch Decision Support Modules VM: PeriWatch Cues™, Curve™ and Vigilance HUB®
  - PeriWatch Tracings Test/Train/Custom Reports VM
- See "Small Site configuration" section for sites <1000 births annually and <10 monitored locations
- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined at system implementation planning stage
- Physical servers (if required) must follow custom/sizing configuration by PeriGen.
- See certificate requirement section for secure system configuration (default)

PeriWatch Tracings – Enterprise Main DB/Application Server VM

# of Monitored locations (1)	50	100	200
# of vCPUs	4	8	8
GB RAM	12	16	20
Volume C: OS	60 GB	60 GB	60 GB
Volume D: Applications	40 GB	40 GB	40 GB

SQL storage requirement when hosted on the local server:

Volume E: SQLData	260 GB	330 GB	400 GB
Volume F: SQLTransactionLog	50 GB	50 GB	50 GB
Volume G : SQLBackup	260 GB	330 GB	400 GB

Bandwidth and performance requirements If SQL server is hosted on a separate cluster or farm:

Transactions / sec	10-50	20-100	40-200
Disk Reads (MMB/sec)	2-60	4-120	8-240
Disk Writes (MB/sec)	1-20	2-40	5-80
Database Size G	260 GB	330 GB	400 GB
Transaction Logs	50 GB	50 GB	50 GB

(1) The CPU necessary on the Server comes from estimating the number of workstations simultaneously using the PeriWatch Tracings Client. This number is often difficult to compute. Monitored locations are easier to determine thus the table above shows Monitored locations. However, it assumes that each Monitored location is running a PeriWatch Tracings Client and there are no more than 10% additional workstations simultaneously doing the same. If it is anticipated that this will not be the case, then it will become necessary to adjust the CPU resources accordingly.

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- Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
- Microsoft Windows 2012 R2 or Windows Server 2016 (2008R2 is no longer supported)
    - Internet Information Services (IIS) installed
    - ASP.NET
    - Windows Authentication
- MS SQL 2014 SP2 or 2016 SP2 (Standard edition or higher), all with Reporting Services (clustered or farm setup supported according to PeriGen guidelines)
- Microsoft Web Services Enhancements (WSE) 3.0
- VC++ Redistributable: 2010 SP1, 2013, 2015
- .Net 3.5 SP1, .Net 4.7.1

PeriWatch Decision Support Server – Enterprise Server VM

	Cues and Curve only			Cues, Curve and HUB		
# of Monitored locations	50	100	200	50	100	200
# of vCPUs	2	4	4	4	6	8
GB RAM	8	12	16	8	12	16
Volume C: OS	40 GB	40 GB	40 GB	40 GB	40 GB	40 GB
Volume D: Applications/SQL	100 GB	100 GB	100 GB	100 GB	100 GB	100 GB

- Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
- Microsoft Windows 2012 R2 or Windows Server 2016 (Windows 2008 R2 is no longer supported)
    - Internet Information Services (IIS) installed
    - ASP.NET
    - Windows Authentication
- MS SQL 2014 SP2 or 2016 SP2 (Standard edition or higher), all with Reporting Services (clustered or farm setup supported according to PeriGen guidelines). Using the PeriWatch Tracings instance of SQL server is not recommended due to performance concerns of the surveillance system.
- VC++ Redistributable: 2010 SP1, 2013, 2015
- .Net 3.5 SP1, .Net 4.7.1 or higher
- .Net Core Windows Hosting 2.2.0

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PeriWatch Tracings - Test/Training/Custom Reports VM

- A minimum of 1 TEST/Train server is mandatory. More servers can be instantiated as needed
- 4 vCPUs
- 8GB RAM
- Storage requirements per volume:
    - C: Operating system - 60 GB
    - D: Applications/SQLData/Logs - 200 GB
- Gigabit Ethernet
- Microsoft Windows 2012 R2 or Windows Server 2016 (Windows 2008 R2 is no longer supported)
    - Internet Information Services (IIS) installed
    - ASP.NET
    - Windows Authentication
- MS SQL 2014 SP2 or 2016 SP2 (Standard edition or higher), all with Reporting Services
- Microsoft Web Services Enhancements (WSE) 3.0
- VC++ Redistributable: 2010 SP1, 2013, 2015
- .Net 3.5 SP1, .Net 4.7.1

Certificate Requirements

- The customer must provide a server-side SSL certificate to encrypt all data in transit across the system.
- The customer is responsible for all certificate renewal management.
- A certificate for SQL TDE is optional for full DB encryption on SQL Enterprise or Data Center versions.
- The SSL certificate requirements are:

Signature Algorithm:	SHA256RSA
Signature hash algorithm:	SHA256
Subject:	<FQDN of server> (ex: VigilanceApp01.Hospital.Org)
Subject Alternative Name:	<FQDN of server> (ex: VigilanceApp01.Hospital.Org)
Public Key:	RSA (2048 Bits)
Key Usage:	Digital Signature, Key Encipherment (a0)
Enhanced Key Usage:	Server Authentication (1.3.6.1.5.5.7.3.1) Client Authentication (1.3.6.1.5.5.7.3.2)

For SQL TDE, the certificate must have the AT_KEYEXCHANGE set to 1

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Small Site Configuration (<1000 births annually and <10 monitoring locations)

- A small site configuration application to <1000 birth and under 10 monitored locations
- With such a small load, all application modules can be consolidated on the primary Application/DB production server
- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined at system implementation planning stage
- If required, physical server(s) option should be equivalent to the VM specifications below.
- Any deviation from the Small Site definition above will require a custom sizing by PeriGen.
- A test server is still mandatory for any configuration

PeriWatch Tracings – Small Site Configuration VM Specification

# of Monitored locations	10
# of vCPUs	4
GB RAM	12
Volume C: OS	60 GB
Volume D: Applications	40 GB
Volume E: SQLData	300 GB
Volume F: SQLTransactionLog	50 GB
Volume G : SQLBackup	250 GB

- Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
- Microsoft Windows 2012 R2 or Windows Server 2016 (Windows 2008 R2 is no longer supported)
    - Internet Information Services (IIS) installed
    - ASP.NET
    - Windows Authentication
- MS SQL 2014 SP2 or 2016 SP2 (Standard edition or higher), all with Reporting Services
- VC++ Redistributable: 2010 SP1, 2013, 2015
- .Net 3.5 SP1, .Net 4.7.1

Virtual Servers

- VMware (6.X and above) and Hyper-V is currently supported for all the configurations above as long as it is part of an enterprise virtualization infrastructure (not on standalone server).
- Other platforms will require extra validation services

Archive Storage

- SAN or NAS storage can be used for long term encrypted archive storage (customer supplied)

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- Archive storage requirements is based on storage requirement of 5 MB per birth.
- Illustration:
    - 8000 births annually for the entire enterprise
    - 8000 births per year x 5MB = 40GB/year.

Client / Workstation Requirements

- Microsoft Windows 7 or 10 Professional 32/64-bit (SP1)
- 2GB RAM. 250 MB of available memory should be reserved for PeriWatch Tracings modules.
- 1280x1024 (4:3) is the recommended standard minimal resolution supported by the system.
- 1024x768 (4:3) is supported with PeriWatch but not supported when Cues & Curve modules are purchased.
- Higher resolutions in 16:9 or 16:10 aspect rations are supported only in native resolution (non-native resolutions or aspect ratios will result in a stretched representation of fetal tracings and are not supported)
- .NET Framework 3.5 Service Pack 1
- .NET Framework 4.7.1 or higher Full client profile
- Microsoft Visual C++ Redistributable 2010 SP1 (x86/x64).
- For Vigilance HUB Admin client: Microsoft Visual C++ Redistributable 2013
- Microsoft Web Services Enhancements (WSE) 3.0
- Wireless network is not recommended for the PeriWatch Tracings application. If required, PeriGen will collaborate with the customer to confirm that PeriWatch application performs as expected on the site's wireless environment.
- Large Screen displays of non-interactive multi-tracing Surveillance or Census screens are supported in full screen in native resolutions only (typically 1920x1080). Different resolutions may require separate validation procedures and must be verified in advance with PeriGen personnel
- IMPORTANT: Interaction and acknowledgement with Fetal Alerts must be done on designated workstations where their MAC address is fixed and mapped in the PeriWatch administrative configuration. These stations are designated "Alerting Stations" and must typically be fixed thick clients.

Thin-Client Technologies

- Citrix or VMware View deployments of PeriWatch Tracings client application is now supported for all stations providing station hostname running the thin-client instance is uniquely identified for each session. Using the server names or synthetic host names is not supported.
- It is the customer responsibility to map the audio channel to the host workstation such that audible fetal alerts can be heard at the point of care.
- Citrix server / VMware view server sizing should be aligned with PeriWatch Client requirements and number of concurrent session requirements (see previous section). Precise server and memory recommendations are a function of overall site load. Recommendations will be finalized during system implementation.

Network (For Workstations and Data Acquisition)

- Switched Ethernet network, 100 Mbit or better

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- CAT5+ dedicated cabling between each monitoring area and networking closet. Point-to-point configuration recommended.
- Proprietary cables between the fetal monitor and wall-plate are provided by PeriGen.

Tracing Data Acquisition

PeriGen provide a proprietary, secure and resilient solution dedicated to the store and forward of tracing data across the system:

- PeriWatch DAS™, a 16 port RS485 to Ethernet device with internal cache. Total number of DAS devices should be determined by total number of monitoring locations, physical vicinity to the networking closets and a 75% port utilization for port redundancy.
- PeriWatch Single gang wall-plate – For bedside wall mounted data collection from a single monitor (with Ethernet pass-through). Typically, one wall-plate is required per monitoring location.
- PeriWatch Dual gang wall-plate – For bedside wall mounted data collection from up to two fetal monitors (with Ethernet pass-through). Dual gang wall-plates are optional.

Fetal Monitor Compatibility

Vendor	Protocol compatibility	Type	Interface card / Firmware	Description
Philips	1371	* 1350A and B * 50XM series	M1350-66532 or M1350-66536 * Firmware: **C01.03 and above**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
Philips	1371	Avalon FM20 Avalon FM30 Avalon FM40 Avalon FM50	Firmware: **F.0X and above.**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
GE/Corometrics	1371	118	J09-J10	DB9 RS232 serial interface
GE/Corometrics	1371	120/170/250 Series	J09-J10-J11	RJ11 RS232 serial interface

* Philips 1350 A/B and 50XM are presently out of support by Philips. Although the serial protocol is still supported, PeriGen cannot be held responsible for known issues with these fetal monitors with firmware versions lower than **C01.03.** Known issues include a very high frequency of SP02 measurements sent to the electronic tracing when the paper recorder door is open, and no paper is in the-0 tray.

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DAS Specifications

Processor Subsystem: Intel® Atom E3815, Single Core

Mainboard: ADLINK Q7-BT1-2G-8G-ER Qseven module

Memory Slot: 2 GB DDR3L

Input / Output:

- Two RS-232 ports (DB9 Adapters)
- Two USB 2.0 ports
- HDMI video interface
- 16 RS-485 ports with RJ45 connectors and +24VDC power for peripherals
- Externally Accessible CFAST Socket

Storage: 8GB SLC industrial temperature CFast card (includes CE core license)

Network Interface: 1 Gigabit Ethernet Port

Power: 100 – 240 VAC, 60 Hz, 2 A input

Operating Temperature: 0 – 40 Degrees Celsius

Enclosure:

- 1U, 19” rack mount compatible enclosure
- Power LED located on front chassis
- Dimensions: 16.75" L x 9.0" W x 1.75" H (19.0" L x 9.0" W x 1.75" H w/ ears)

Warranty: Three years from date of shipment

Regulatory:

- UL 60950-1/R:2014-10
- CAN/CSA C22.2 No. 60950-1/A2:2014
- EN: 55024:2010
- EN: 55032:2012
- EN: 61000-3-3:2013
- FCC, 47 CFR, Part 15 Subpart B
- ISED Canada ICES-003 Issue 6

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Network ports used by the system and accessories devices

	Direction relative to device	Source	Destination	Protocol	App Protocol	Port	PHI	Description
Core system services	Bi-directional	DAS, Workstations	App Server	TCP	HTTPS	443	Yes/ Secure	Mandatory. All Communications with the Repository back-end
	Bi-directional	DAS	App Server	TCP	Proprietary	8000	No	Mandatory. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Workstations	App Server	TCP	Proprietary	8002	No	Mandatory. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Workstations	App Server	TCP	Proprietary	8004 (See 8014)	Yes	Mandatory. Send / Receive Messages from Clients to be broadcast to Workstations
	Bi-directional	Workstations	App Server	TCP	Proprietary	8014	Yes / Encrypted	Mandatory. Send / Receive Messages from Clients to be broadcast to Workstations
	Bi-directional	Workstations	App Server	TCP	HTTP	8005	Yes	Only used by the Newborn Module
	Bi-directional	Workstations	App Server	TCP	Proprietary	8006	Yes / Secured	Only used by the Newborn Module
	Bi-directional	Workstations	App Server	TCP	Proprietary	8007	Yes / Secured	Only used by the Newborn Module
	Bi-directional	Workstations	App Server	TCP	HTTPS	8008	Yes / Secured	Only used by the Newborn Module
	Bi-directional	Workstations	App Server	TCP	Proprietary	8009	No	Only when using/enabling Single-Sign-On
	Bi-directional	DAS, Workstations	App Server	UDP	Proprietary	11002	No	Time synchronization to the main app server
Ports used when Patterns/Cues are installed	Bi-directional	Decision Support Server	App Server	TCP	Proprietary	7101	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Data communication to the decision support modules
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7802	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Used for tracings interpretation data feed
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7803	Yes /Encrypted	Only when Cues/Patterns is purchased/installed. Plugins service, Cues algorithm data feed
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8081	Yes / Secured	Only when Cues/Patterns is purchased/installed. Data feed from decision support modules
	Bi-directional	Workstations	Decision Support Server	TCP	SignalR	8082	No / Secured	Only when Cues/Patterns is purchased/installed. Live data feed from decision support modules
Ports used when the Vigilance Platform is installed	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8083	Yes / Secured	Only when Vigilance is purchased/installed. Vigilance dashboard – single patient browser-based app
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8084	Yes / Secured	Only when Vigilance is purchased/installed. Vigilance HUB multi-patient browser-based app
	Bi-directional	Decision Support Server	Decision Support Server	TCP	Net.tcp	8201	Yes / Secured	Only when Vigilance is purchased/installed. HUB DataCollector, patient information & Cues algorithm data feed
	Bi-directional	Decision Support Server	Decision Support Server	TCP	Net.tcp	8203	Yes / Secured	Only when Vigilance is purchased/installed. HUB DBService, patient information & Cues algorithm data feed

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Ports used when DIGI/MOXA terminal servers are used (typically only with Vigilance standalone system)	Bi-directional	Network Redirectors	App Server	TCP	Proprietary	8012	No	ONLY IF DIGI SERIAL SERVERS ARE USED. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Moxa Nport 6000 Series	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
	Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	HTTP	80	No	Device configuration web UI
	Bi-directional	Moxa Nport 6000 Series	App Server	TCP	HTTPS	443	No	Secure Device configuration web UI
	Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	Proprietary	950-965	No	Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	Proprietary	966-981	No	Command Port
	Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	UDP	Proprietary	4800	No	Broadcast search, Monitoring the status of the NPort, getting the current settings, and Real COM mode mapping
	Bi-directional	Moxa Nport 5000 Series	App Server	TCP	Proprietary	4900	No	Device Firmware Upgrade Port
	Bi-directional	Moxa Nport 6610-32	App Server	TCP	Proprietary	982-997	No	Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Moxa Nport 6610-32	App Server	TCP	Proprietary	998-1013	No	Command Port
	Bi-directional	Digi	App Server	TCP	Proprietary	771	No	RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Digi	App Server	TCP	Proprietary	1027	No	Secure RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Digi	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
	Bi-directional	Digi	App Server	TCP	HTTP	80	No	Device configuration web UI

Note: The above list outlines all possible ports with the PeriWatch system (across all possible modules). Always confirm with the PeriGen technical consultant on the applicable ports for your precise configuration.

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